

ASTRA — Complete Technical Product Flow Document

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Audience: Backend engineers, API contract designers, Flutter developers
Source: App screenshots (Polar + Non-Polar flows) + architecture.md

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1. Screen Inventory

1.1 Splash Screen

Screen Name: `SplashScreen`

Route: `/` (initial)

Purpose: App entry point. Checks auth state and onboarding completion, then routes appropriately.

Component	Type	Notes
ASTRA logo (A)	Image	Centered, teal gradient
"ASTRA" wordmark	Text	White, spaced caps
"YOUR PERSONAL INTELLIGENCE PLATFORM"	Text	Muted, small caps
Progress dots (partial)	Indicator	Loading state

Navigation:

- Auth token valid + onboarding complete → `/home`
- Auth token valid + onboarding incomplete → `/onboarding/step/{n}`
- No auth token → `/welcome`

State Checks (SharedPreferences):

- `auth_token` — present → proceed to onboarding check
- `onboarding_complete` — bool

- **onboarding_step** — int (resume partial onboarding)

Loading State: Full-screen logo display (minimum 1.5s, then routes)

Error State: Network timeout is ignored — routing decision is local only

1.2 Welcome / Onboarding Carousel

Screen Name: **WelcomeScreen**

Route: **/welcome**

Purpose: Marketing intro with 3 carousel slides. Entry point for Sign Up and Log In.

Component	Type	Notes
ASTRA logo	Image	Top center
"Mental Performance AI"	Subtitle	Below logo
Carousel slide image	Image	Full-width card with rounded corners
Slide title "Performance"	Text	Bold, below image
Slide subtitle	Text	"Track mental performance, recovery and readiness..."
Dot indicators	Row	3 dots, current page highlighted
"Sign Up Now" button	Primary CTA	Teal filled, full width
"Log In Here"	Text link	Teal color, below button

Carousel Slides (inferred from dots):

1. Performance — "Track mental performance, recovery and readiness to unlock your full potential in competition."
2. [Needs backend confirmation — slide 2 content not captured]
3. [Needs backend confirmation — slide 3 content not captured]

Navigation:

- "Sign Up Now" → **/signup**
- "Log In Here" → **/login** (same flow, different label — **Needs backend confirmation**)

Empty/Error State: Static — no API calls on this screen.

1.3 Sign Up Screen

Screen Name: **SignUpScreen**

Route: **/signup**

Purpose: Entry method selection for new user registration.

Component	Type	Notes
Back arrow (<)	Icon button	→ /welcome

Component	Type	Notes
"Sign Up"	AppBar title	
"Meet Astra"	Heading	
Subtitle	Text	"Understanding what drives outcomes—from performance to relationships to daily life."
"Continue" button	Primary	→ /phone-entry (phone OTP flow)
"OR" divider	Text	
"Continue with Google"	Outlined button	Google icon + text
"Continue with Apple"	Outlined button	Apple icon + text
Privacy Policy link	Tap link	External URL
Privacy disclaimer	Text	Small, below buttons

Navigation:

- "Continue" → PhoneEntryScreen
- "Continue with Google" → Google OAuth flow (**Needs backend confirmation** — Supabase Google OAuth)
- "Continue with Apple" → Apple OAuth flow (**Needs backend confirmation** — Supabase Apple OAuth)

Validation: None on this screen (selection only).

1.4 Phone Number Entry Screen

Screen Name: PhoneEntryScreen

Route: /phone-entry

Purpose: Collect mobile number for OTP-based authentication.

Component	Type	Notes
Back arrow	Icon button	→ /signup
"Mobile number verification"	AppBar title	
Phone illustration	Image	
"Enter your phone number"	Heading	
"Mobile number"	Label	

Component	Type	Notes
Country picker "IN (+91)"	Dropdown	Country code selector
Phone input	TextField	Placeholder "123-456-7890", numeric keyboard
Hint text	Text	"ASTRA will send you a text..."
"What if my number changes?"	Text link	Needs backend confirmation
"Continue" button	Primary	Disabled until phone entered; → OTP screen

Validation:

- Phone number: 10 digits minimum (post country code strip)
- Country code: must be selected
- Button disabled until valid format

Loading State: Button shows spinner while Supabase OTP request is in flight.

Error States:

- Invalid phone format → inline error below input
- Supabase OTP failure → toast "Failed to send code. Try again."
- Rate limit hit → toast "Too many attempts. Wait 60 seconds."

API Call:

```
POST /auth/v1/otp
Body: { "phone": "+919999999999" }
```

1.5 OTP Verification Screen

Screen Name: `OtpVerificationScreen`

Route: `/otp-verify`

Purpose: Verify 6-digit OTP sent to the user's mobile number.

Component	Type	Notes
Back arrow	Icon button	→ <code>/phone-entry</code>
"Mobile number verification"	AppBar title	
Envelope illustration	Image	
"Enter your verification code"	Heading	
6 OTP input boxes	PinInput	Auto-advance on digit entry
"Sent to 911-472-5836. Edit"	Text	"Edit" is a tap link → back to phone entry
"Didn't get a code?"	Text link	Triggers resend OTP

Component	Type	Notes
"Continue" button	Primary	Disabled until 6 digits entered; verifies OTP

Validation:

- Exactly 6 digits
- Auto-submit on 6th digit entry

Loading State: Button shows spinner during verification.

Error States:

- Wrong OTP → "Invalid code. Please try again."
- Expired OTP → "Code expired. Tap 'Didn't get a code?' to resend."
- Max attempts → "Too many attempts. Request a new code."

On Success:

- First-time user → `/onboarding/step/1`
- Returning user (onboarding complete) → `/home`
- Returning user (partial onboarding) → `/onboarding/step/{saved_step}`

API Calls:

```
POST /auth/v1/verify
Body: { "phone": "+919999999999", "token": "123456", "type": "sms" }

Response (success):
{
  "access_token": "<jwt>",
  "refresh_token": "<opaque>",
  "user": { "id": "<uuid>", ... }
}
```

SharedPreferences Written:

- `auth_token`
- `refresh_token`

1.6 Onboarding Step 1 — Personal Details

Screen Name: `OnboardingStep1Screen`

Route: `/onboarding/step/1`

Purpose: Collect basic personal information to create athlete profile.

Component	Type	Notes
Back arrow	Icon button	→ previous or Welcome

Component	Type	Notes
Progress bar	LinearProgress	1/4 filled
"STEP 1 OF 4"	Caption	
"Welcome to Astra Performance."	Heading	Bold
Subtitle	Text	"Let's build your baseline..."
"Full Name" label + input	TextField	Placeholder "e.g. Jane Doe"
"Age" label + input	TextField	Numeric, placeholder "Enter your age"
"Gender"	Label	
Male / Female / Other	Checkboxes	Single-select (radio behavior)
"City" label + input	TextField	Placeholder "Where do you live?"
"Continue to Step 2"	Primary button	Disabled until all required fields filled

Validation:

- Full Name: required, min 2 chars
- Age: required, numeric, 10–80 range
- Gender: required (one selection)
- City: required, min 2 chars

Loading State: None (local step)

Navigation: → /onboarding/step/2

SharedPreferences Written: onboarding_step = 1, partial profile data

1.7 Onboarding Step 2 — Shooting Profile

Screen Name: OnboardingStep2Screen

Route: /onboarding/step/2

Purpose: Collect sport-specific profile to personalize the training dashboard.

Component	Type	Notes
Back arrow	Icon button	→ Step 1
Progress bar	LinearProgress	2/4 filled
"STEP 2 OF 4"	Caption	
"Your shooting profile"	Heading	
Subtitle	Text	"Tell us about your current discipline..."

Component	Type	Notes
"Discipline" dropdown	DropDownButton	Default "Air Pistol"; options: Air Pistol, Air Rifle, 10m, 25m Pistol, 50m Rifle, etc. (Needs backend confirmation for full list)
Helper text	Text	"select your primary shooting event"
"Experience Level"	Label	
Beginner / Intermediate / Advance	Segmented control	Single-select
"Years shooting"	Label	
Stepper (– / value / +)	Counter	Default 0, min 0
"Academy / Club" label	Label	Marked "optional"
Academy input	TextField	Placeholder "Enter your academy or club"
"Continue to Step 3"	Primary button	

Validation:

- Discipline: required (dropdown selection)
- Experience Level: required
- Years shooting: >= 0
- Academy: optional

Navigation: → </onboarding/step/3>

SharedPreferences Written: [onboarding_step](#) = 2

1.8 Onboarding Step 3 — Performance Baseline

Screen Name: [OnboardingStep3Screen](#)

Route: </onboarding/step/3>

Purpose: Capture current performance level and identify mental obstacles.

Component	Type	Notes
Back arrow	Icon button	→ Step 2
Progress bar	LinearProgress	3/4 filled
"STEP 3 OF 4"	Caption	

Component	Type	Notes
"Where are you now?"	Heading	
Subtitles	Text	"Let's understand your baseline..."
"Average practice score:"	Label	
Score input	TextField	Numeric, placeholder "150"
"Target score:"	Label	
Target input	TextField	Numeric, placeholder "580"
"What's getting in the way?"	Heading	"Select up to 3"
Checkbox list items	Checkboxes	5 options (multi-select up to 3):
• Nervousness in competition	Checkbox	
• Overthinking scores	Checkbox	
• Poor sleep	Checkbox	
• Distractions	Checkbox	
• Pressure in big moments	Checkbox	
"Continue to Step 4"	Primary button	

Validation:

- Average practice score: numeric, optional
- Target score: numeric, optional
- Obstacles: 0–3 selections (enforce max 3)

Navigation: → /onboarding/step/4

SharedPreferences Written: baseline_scores, onboarding_step = 3

1.9 Onboarding Step 4 — Goal Setting

Screen Name: OnboardingStep4Screen

Route: /onboarding/step/4

Purpose: Set 30-day and 6-month performance targets.

Component	Type	Notes
Back arrow	Icon button	→ Step 3
Progress bar	LinearProgress	4/4 filled
"STEP 4 OF 4"	Caption	
"Set your goals"	Heading	

Component	Type	Notes
Subtitle	Text	"Specific goals give your training direction..."
"30-day target" label	Label	
Duration chip "Next 30 days"	Chip	Non-interactive label
30-day textarea	TextField multiline	Placeholder "Example: Average 575+ in competition"; mic icon for voice input
"6-month target" label	Label	
Duration chip "6 months"	Chip	Non-interactive label
6-month textarea	TextField multiline	Placeholder "Example: Compete at nationals with 590+ average"; mic icon
Tip box	Card	Yellow border, "Best goals are specific and measurable..."
"Continue" button	Primary	
"You can update your goals anytime"	Caption	Below button

Validation:

- Both fields: optional but encouraged

Navigation: → Psychology Questions (</psychology/questions>)

SharedPreferences Written: [onboarding_step](#) = 4

API Call (on Continue):

Persist athlete profile to Supabase — all 4 steps batched:

```
POST /rest/v1/athletes
Headers: Prefer: return=representation
Body: { full_name, date_of_birth (inferred from age), gender, ... }

POST /rest/v1/athlete_details
Body: { athlete_id, discipline, experience_years, goals: [goal_30d, goal_6m], ... }
```

1.10 Psychology Assessment Questions

Screen Name: [PsychologyQuestionsScreen](#)

Route: </psychology/questions>

Purpose: Collect baseline psychology responses across dimensions (25 questions total).

Component	Type	Notes
"Questions" label	AppBar label	Left-aligned
"2/25" counter	Chip	Top-right, shows progress
Progress bar	LinearProgress	Top of screen, thin
Question text	Heading large	Scenario-based question
Answer choices	Checkbox list	4 options, single-select
"Explain your answer" label	Text	Teal color
Explanation textarea	TextField multiline	Placeholder "Write your message here..."
"Back"	Text button	Left
"Next"	Primary button	Right; advances to next question
"Skip for now"	Text link	Below buttons; skips entire section

Sample Question (visible):

"You have hit a string of perfect 10s and feel a 'rush' of excitement. How do you handle this sudden surge?"

- Acknowledge the success, exhale, and return to my technical process.
- Enjoy the feeling but try to keep my physical movements steady.
- **Start thinking about the perfect score and feel my heart race.** (selected)
- Get overexcited and rush the next shot to keep the streak going.

Validation:

- Selection required before Next (or use Skip)
- Explanation: optional free text

Navigation:

- "Next" on last question → Polar Permissions screen
- "Skip for now" → Polar Permissions screen
- Incomplete assessment → shown as "Psychology Assessment Pending" on Home dashboard

API Call (per answer):

```
POST /rest/v1/psychology_responses
Body: {
  "athlete_id": "<uuid>",
  "question_id": "<uuid>",
  "response_value": "3",           // option index or text
  "explanation": "optional text"   // if provided
}
```

Database Read: `SELECT * FROM psychology_questions ORDER BY display_order`

Empty State: If no questions configured → skip screen (**Needs backend confirmation**)

1.11 Polar Permissions Screen

Screen Name: PolarPermissionsScreen

Route: /polar/permissions

Purpose: Request required system permissions before Polar BLE setup.

Component	Type	Notes
Heart icon	Image	Pink/red square, top-left
"Your heart tells the truth"	Heading	
Subtitle	Text	"We use your Polar device to measure stress, focus, and recovery..."
"To continue, we need"	Divider label	
Bluetooth row	ListTile	Bluetooth icon, "To connect to Polar", chevron → system dialog
Notifications row	ListTile	Bell icon, "For session reminders", chevron → system dialog
Microphone row	ListTile	Mic icon, "Optional" chip, "For session reminders", chevron
"Continue & Allow"	Primary button	Triggers permission requests
"I don't have a polar yet"	Text link	→ skips to /home without BLE setup

Navigation:

- "Continue & Allow" → triggers BLE scan → PolarScanScreen
- "I don't have a polar yet" → /home (Polar-not-connected state)

Permission Logic (Android/iOS):

1. Bluetooth scan + connect permissions
2. Push notification permission
3. Microphone (optional, skip if denied)

1.12 Polar Device Scan Screen

Screen Name: PolarScanScreen

Route: /polar/scan

Purpose: Scan for nearby Polar devices and allow user to select and connect.

Component	Type	Notes
WiFi-like scan icon	Image	Teal circle

Component	Type	Notes
"Can't find your device?"	Text	Above device list when no device found
"Try again or continue without Polar for now."	Subtitle	
"Having trouble?"	Section heading	
"No Polar device was found during this scan."	Body	
"Found devices" divider	Label	
Device list item	ListTile	Bluetooth icon, "Polar H10 (Demo)", "H10" subtitle, "Connect" button (teal)
Warning card	Card	Orange border, "Having trouble?" heading, explanation text, "Troubleshoot / Try again" teal link
"Continue without Polar"	Primary button	Bottom → /polar/baseline or /home

States:

- Scanning: pulsing icon animation, no devices listed yet
- Device found: device list appears with "Connect" buttons
- Device not found after timeout: warning card shown
- Connected: → transition to [PolarConnectedScreen](#)

Interactions:

- "Connect" button on device → initiates [PolarBleApi.connectToDevice\(deviceId\)](#)
- "Troubleshoot / Try again" → retriggers scan
- "Continue without Polar" → [/home](#) (Polar-not-connected state)

1.13 Polar Connected Confirmation Screen

Screen Name: [PolarConnectedScreen](#)

Route: [/polar/connected](#)

Purpose: Confirm successful device connection before proceeding to baseline measurement.

Component	Type	Notes
Green checkmark circle	Image	Green background
"Connected to Polar H10 (Demo)"	Caption	Green text
"Device connected"	Heading	Bold

Component	Type	Notes
Body text	Text	"Your heart rate and HRV data will now be tracked during sessions."
"Found devices" divider	Label	(visible but list empty — all connected)
"Done Let's go"	Primary button	→ BaselinePromptScreen

SharedPreferences Written: [polar_device_id](#) = deviceId

API Call:

```
POST /rest/v1/polar_devices
Headers: Prefer: return=representation
Body: {
  "athlete_id": "<uuid>",
  "device_id": "Polar_H10_Demo",
  "model": "H10",
  "last_connected": "<now_utc>",
  "is_primary": true
}
```

1.14 Baseline Measurement Prompt Screen

Screen Name: [BaselinePromptScreen](#)

Route: [/polar/baseline/prompt](#)

Purpose: Instruct athlete to sit still for 60-second resting HR baseline measurement.

Component	Type	Notes
Heart icon	Image	Pink/red square
"Let's measure your baseline"	Heading	
Body text	Text	"We need to know your resting heart rate..."
"Sit comfortably" row	ListTile	Bluetooth icon (repurposed)
"Breathe normally" row	ListTile	Bell icon
"Don't move" row	ListTile	Mic icon
"Start - 60 sec"	Primary button	→ BaselineCapturingScreen

Navigation: "Start - 60 sec" → begins 60-second HR capture

1.15 Baseline Capturing Screen

Screen Name: BaselineCapturingScreen

Route: /polar/baseline/capturing

Purpose: Live 60-second resting HR measurement with countdown and waveform visualization.

Component	Type	Notes
"Capturing baseline..."	Body text	Top
Heart animation	Pulsing circle	Pink circle with heart icon
BPM value "75"	Large text	Teal/brand color, updates live
"BPM" label	Caption	
HR waveform	Bar chart	Red bars, last ~30 readings
"01:00 remaining"	Timer	Clock icon + countdown
Progress bar	LinearProgress	Teal, advances as time counts down
"Stay still - movement affects accuracy"	Caption	Below progress bar

Behavior:

- Subscribes to HR stream from Polar H10
- Accumulates 60 readings
- Countdown from 60 → 0
- On completion → calculates mean BPM → BaselineResultScreen
- If Polar disconnects during capture → show error, offer retry

1.16 Baseline Result Screen

Screen Name: BaselineResultScreen

Route: /polar/baseline/result

Purpose: Show captured resting HR and confirm baseline saved.

Component	Type	Notes
Green checkmark circle	Image	
"Your resting heart rate is"	Heading	
BPM value "66"	Large text	Green, bold
"BPM" label	Caption	
"Baseline saved" card	Card	Green border, explanatory text: "This is your personal baseline. We'll use it to detect when stress affects your performance..."

Component	Type	Notes
"Continue"	Primary button	→ <code>OnboardingCompleteScreen</code>

SharedPreferences Written: `baseline_hr` = 66 (int)

API Call:

```
PATCH /rest/v1/athlete_details?athlete_id=req.<uuid>
Body: { "resting_hr": 66 }
```

1.17 Onboarding Complete / You're All Set Screen

Screen Name: `OnboardingCompleteScreen`

Route: `/onboarding/complete`

Purpose: Summarize onboarding results and launch into the main app.

Component	Type	Notes
WiFi/connected icon	Image	Teal circle
"You're all set, {first_name}"	Heading	Personalized
"Here's your starting point:"	Subtitle	
Summary card	Card	3 rows: Discipline / Goal / Resting HR
Discipline row	Row	"Air Pistol"
Goal row	Row	"560 in 30d"
Resting HR row	Row	"68 bpm"
"Ready to train"	Caption	
"Go to Home"	Primary button	→ <code>/home</code>

SharedPreferences Written:

- `onboarding_complete` = true
- `onboarding_step` = 4

1.18 Athlete Home Screen (Dashboard)

Screen Name: `AthleteHomeScreen`

Route: `/home`

Purpose: Main dashboard showing daily readiness metrics, session CTA, and AI coach summary.

TWO STATES:

State A — Polar Connected:

Component	Notes
ASTRA logo + "Athlete Home"	Header
Bell icon	Notification button, top-right
"Polar connected" badge	Green dot + teal chip — Polar connected state
Psychology Assessment Pending card	"You have 23 questions remaining" + "Continue" teal button → /psychology/questions
READINESS card	"60" + "moderate" label
RECOVERY card	"48" + "acceptable" label
STRESS card	"40" + "moderate" label
MENTAL STATE card	"Steady" (text value) + "steady" label
Session Action Center card	"Start Training Session" button → SessionSetup
Coach Quick Summary card	Numbered 1-2-3 list of coach tips + Saarthi avatar
Bottom nav	Home (active) / Sessions / Insight / Profile

State B — Polar NOT Connected:

Component	Notes
"Polar not connected" card	Heart icon, "Connect Polar to unlock BPM, HRV, and physiology insights." + "Connect Polar Device" button → /polar/scan
All other cards	Same as State A
Saarthi avatar	Floating action area, bottom-right

Readiness/Recovery/Stress/Mental State Computation:

- Derived from last [daily_checkins](#) record
- If no checkin today → values default to 0 or "—"

API Calls (on mount):

```
GET /rest/v1/daily_checkins?athlete_id=eq.<uuid>&checkin_date=eq.<today>&limit=1
GET /rest/v1/ai_insights?
athlete_id=eq.ATH001&is_read=eq.false&order=priority.desc&limit=3
GET /rest/v1/performance_summary?
athlete_id=eq.ATH001&order=created_at.desc&limit=1
POST /rest/v1/rpc/get_dashboard_data body: { "p_athlete_id": "ATH001" }
```


Navigation:

- "Continue" (Psychology pending) → /psychology/questions
- "Start Training Session" → CheckInScreen (pre-session flow starts with check-in)
- "Connect Polar Device" → /polar/scan
- Saarthi avatar → SaarthiChatScreen
- Bell → Notifications (Needs backend confirmation)
- Bottom nav Sessions → /sessions
- Bottom nav Insight → /insights
- Bottom nav Profile → /profile

1.19 Saarthi AI Chat Screen

Screen Name: SaarthiChatScreen

Route: /saarthi

Purpose: Conversational AI coaching interface with quick-action shortcuts.

Component	Type	Notes
Back arrow	Icon button	
Saarthi avatar (circular)	Image	Animated character, blue/teal
"Saarthi AI"	Title	
"● Online"	Status	Green dot
Close (x)	Icon button	Dismiss chat
Talk to Saarthi banner	Card	Large animated banner with "Talk to Saarthi, your personal guide" + waveform animation + chevron
"Quick actions"	Section label	
"Analyze last session"	Quick action card	Icon + title + subtitle "Review shot spread and metrics"
"Technique tips"	Quick action card	"Get advice on grip and breathing"
"Plan next training"	Quick action card	"Set goals for your upcoming session"
"Breathing control"	Quick action card	"Improve your breathing techniques"
"Mental focus coaching"	Quick action card	"Build focus and concentration"
Attachment icon	Icon button	Message input left

Component	Type	Notes
"Type your message..."	TextField	Chat input
Send button	Icon button	Teal circle → sends message

Quick Action Behavior:

Each quick action → pre-fills context message and sends automatically to Saarthi Edge Function.

Message Flow:

- 1. User types or taps quick action
- 2. Message appended to chat history locally
- 3. POST to Saarthi Edge Function
- 4. Response appended to chat history
- 5. INSERT both turns to **athlete_chat** table (non-blocking)

Empty State: Banner + Quick actions grid (no chat history)

Loading State: Typing indicator (3 dots animation) while waiting for Saarthi response

API Call:

```
POST /functions/v1/saarthi-chat
Headers: Authorization: Bearer <user_jwt>
Body: {
  "athlete_id": "<uuid>",
  "session_id": null,
  "message": "Analyze last session",
  "context_mode": "general"
}
Response: {
  "reply": "...",
  "suggestions": [...],
  "mood_flag": null
}
```

1.20 Daily Check-in Screen

Screen Name: **DailyCheckinScreen**

Route: **/checkin**

Purpose: Pre-session emotional and physical readiness check. Feeds dashboard metrics.

Component	Type	Notes
Back arrow	Icon button	
"Daily Check-in"	AppBar title	

Component	Type	Notes
"How are you feeling?"	Section heading	
5 emoji options	Row of cards	Trouble / Poor / Okay / Good / Great (single select)
"Energy level"	Section heading	
Slider	Slider	1–10, Low → High labels, current value "5" shown below
"How did you sleep?"	Section heading	
Sleep duration chips	Row	<5h / 5-6h / 6-7h / 7-8h / 8h+ (single select)
"Any of these apply?" (optional)	Section heading	
Mood tags	Wrap/chips	Anxious / Confident / Focused / Distracted / Motivated / Calm (multi-select)
"Save Check-in and Continue"	Primary button	Disabled until feeling + energy + sleep selected

Validation:

- Feeling: required (one emoji)
- Energy level: required (slider, default 5)
- Sleep duration: required (one chip)
- Mood tags: optional

Mapping (feeling emoji → numeric score):

- Trouble = 1–2
- Poor = 3–4
- Okay = 5–6
- Good = 7–8
- Great = 9–10

API Call:

```
POST /rest/v1/daily_checkins
Headers: Prefer: return=representation
Body: {
  "athlete_id": "<uuid>",
  "checkin_date": "2026-05-19",
  "mood": 7,
  "energy": 5,
  "sleep_quality": 8,
  "sleep_hours": 7.5,    // mapped from chip selection
}
```

```
    "stress_level": null, // not collected here – Needs backend
    confirmation
    "soreness": null,
    "notes": "Anxious, Focused" // comma-joined tags
  }
```

SharedPreferences Written: `current_checkin` = JSON

Navigation: → `SessionSetupScreen`

1.21 Session Setup Screen

Screen Name: `SessionSetupScreen`

Route: `/session/setup`

Purpose: Configure session parameters before beginning training.

Component	Type	Notes
Back arrow	Icon button	
"Set up your session"	AppBar title	
"New session"	Heading	
"Set up your training parameters."	Subtitle	
"Range type today"	Label	
Paper / Electronic toggle	SegmentedButton	Paper (dark/selected) / Electronic
"Session type"	Label	
Scoring / Grouping / Dry Fire	Chip selection	Single-select
"Planned shots"	Label	
Stepper (– / value / +)	Counter	Default 60
Quick-select chips	Row	20 / 40 / 60 (highlighted) / 80 / 100 / 120
"Discipline"	Label	
Discipline dropdown	DropDownButton	Default "Air Pistol" (from profile)
"Begin Ritual"	Primary button	Play icon + text → <code>PreSessionRitualScreen</code>

Validation:

- Range type: required (default Paper)
- Session type: required (no default — must select)
- Planned shots: required (default 60)
- Discipline: required (defaults from profile)

SharedPreferences Written: `session_setup` = JSON {rangeType, sessionType, plannedShots, discipline}

API Call (non-blocking session creation):

```
POST /rest/v1/App_Sessions
Headers: Prefer: return=representation
Body: {
  "athlete_id": "ATH001",
  "session_type": "training",
  "range_type": "paper",
  "planned_shots": 60,
  "start_time": "<now_utc>",
  "status": "active"
}
Response: [{ "id": "<session_uuid>", ... }]
```

SessionMemory Written: `SessionMemory.sessionId` = response id

1.22 Pre-Session Ritual Screen

Screen Name: `PreSessionRitualScreen`

Route: `/session/ritual`

Purpose: Guided breathing/focus routine before session begins. **(Needs backend confirmation** — content not visible in screenshots; inferred from "Begin Ritual" button)

Inferred UI (based on button label + app context):

- Animated breathing guide (expand/contract)
 - Timer countdown
 - Optional Saarthi voice cue
 - "Start Session" button → `LiveTrainingScreen`
-

1.23 Live Training Screen (Polar Connected)

Screen Name: `LiveTrainingScreen`

Route: `/session/live`

Purpose: Active session monitoring with real-time heart rate display and session controls.

Polar-Connected State:

Component	Type	Notes
"● Live" badge	Status chip	Green dot + "Live"
"Paper Session"	AppBar title	Dynamic (Paper/Electronic)
Timer "0:01"	Timer	Session elapsed time, top-right
HR card	Card	Heart icon, "72 bpm", "Baseline 65 ↑ +7"

Component	Type	Notes
HR bar chart	Chart	Red bars, "HR Last 30 Readings" label
"Pause"	Secondary button	Suspends session, telemetry pauses
"✓ End Session"	Primary button	Ends session → ScoreEntryScreen

Non-Polar State:

No HR card — screen may show timer only or go directly to score entry (**Needs backend confirmation**)

Telemetry Behavior:

- HR stream: subscribed → buffered every 10s → bulk POST
- Pause: suspends buffer flush, marks session paused in Supabase
- End: flushes remaining buffer, marks session completed

API Calls (continuous, non-blocking):

```
POST /rest/v1/hr_stream    (every 10s)
Headers: Prefer: return=minimal
Body: [ array of { time, session_id, athlete_id, bpm, rr_interval } ]

PATCH /rest/v1/App_Sessions?id=eq.<session_id>
Body: { "status": "paused" }    // on Pause

PATCH /rest/v1/App_Sessions?id=eq.<session_id>
Body: { "status": "active" }    // on Resume
```

Navigation:

- "End Session" → ScoreEntryScreen (if paper) or directly to ReflectionScreen (if electronic — **Needs backend confirmation**)

1.24 Score Entry Screen

Screen Name: ScoreEntryScreen

Route: /session/score

Purpose: Enter series totals for paper target sessions. Series count derived from plannedShots / 10.

Component	Type	Notes
Back arrow	Icon button	
"Paper Session"	AppBar title	
"Enter your series total score"	Heading	

Component	Type	Notes
"Enter the total score for this series."	Subtitle	
"Series {n} of {total}"	Label	e.g. "Series 2 of 2"
Active series card	Card	Series number, "10 shots", "Max score: 109", input field, "Save Score" button
Score input	TextField	Large numeric input, placeholder "0"
"Save Score ✓"	Button	Inside card, teal
"So far in session"	Section label	
Past series list	List	Series 1: 65, Series 2: — (pending)
"Running total" card	Card	Trend icon, "Running total", "After 1 of 2 series", score "65 / 218"
"↶ Undo last entry"	Text link	Bottom, teal

Max Score Calculation:

```
max_per_series = shots_per_series × 10.9    // 10 shots × 10.9 = 109
total_max = series_count × max_per_series
```

Validation:

- Score: numeric, $0 \leq \text{value} \leq \text{max_per_series}$
- "Save Score" disabled until valid score entered

"All series entered" State:

When all series are filled:

- Header changes to "All series entered"
- Subtitle: "Review your scores and continue."
- "Continue to Summary ✓" button appears (primary)
- All series listed with their scores
- "Running total" shows final total / max

Navigation: "Continue to Summary" → [ReflectionScreen](#)

SharedPreferences Written: `score_summary` = [{seriesNumber, shots, total}, ...]

API Calls (non-blocking, after all series saved):

```
POST /rest/v1/performance_summary
Headers: Prefer: return=representation
```

```
Body: {
  "athlete_id": "ATH001",
  "session_id": "<session_uuid>",
  "total_score": 123,
  "max_possible": 218,
  "avg_per_shot": 6.15,
  "score_variance": 0.34,
  "consistency_pct": 62.0,
  "performance_rating": 6,
  "focus_level": "moderate"
}
```

1.25 Reflection Screen

Screen Name: ReflectionScreen
Route: /session/reflection
Purpose: Post-session debrief with emotional rating and qualitative feedback.

Component	Type	Notes
Back arrow	Icon button	
"Reflect"	Heading	
"A few quick questions to choose the loop"	Subtitle	
"How did it feel?"	Section heading	
5 emoji options	Row	Trouble / Poor / Okay / Good / Great (selected, circled)
"What Worked"	Section heading	
Text area	TextField multiline	"Write your message here..."
"What Didn't"	Section heading	
Text area	TextField multiline	"Write your message here..."
"What went well today?"	Section heading	
Text area	TextField multiline	"Text area for athlete input"
"Save & View Summary"	Primary button	

Validation:

- Emoji rating: required
- Text areas: optional

API Calls:

```
POST /rest/v1/session_reflections
Headers: Prefer: return=representation
Body: {
  "session_id": "<session_uuid>",
  "athlete_id": "<uuid>",
  "what_went_well": "...",
  "what_to_improve": "...",
  "key_takeaway": "...",
  "emotion_rating": 5
}

PATCH /rest/v1/App_Sessions?id=eq.<session_uuid>
Body: { "status": "completed", "end_time": "<now_utc>" }
```

Navigation: → [SessionSummaryScreen](#)

1.26 Session Summary Screen

Screen Name: [SessionSummaryScreen](#)

Route: [/session/summary](#)

Purpose: Post-session performance overview with metric cards, history, and score analysis.

Component	Type	Notes
Back arrow	Icon button	
"Astra Performance"	Supertitle	Teal
"Sessions"	Heading	
"Last session summary, history and score analysis"	Subtitle	
PERFORMANCE card	MetricCard	"6.2" + "● last session"
READINESS card	MetricCard	"62" + "● warm up slowly"
HOLD STABILITY card	MetricCard	"6.5" + "● session best"
MENTAL SCORE card	MetricCard	"55" + "● steady focus"

Component	Type	Notes
"Last Session Summary" card	Card	Teal headline "Stable hold with recovery watch", "Avg 6.2/shot · 20 shots · 0:00", "Best series: S1: 65 · Worst: S2: 58"
"Session History" card	Card	Latest row: Score 6.2 / Eff 62% / Best S1:65; Earlier row: Worst S2:58 / Shots 20 / 0:00
"Score Analysis" card	Card	"Avg 6.2/shot · efficiency at 62%.", "Best shot: 6.5 · Worst: 5.8 · Consistency gap: 0.7 pts."
"Save & View Full Report"	Primary button	→ SessionReportScreen

Metric Definitions:

- PERFORMANCE = performance_rating (avg/shot)
- READINESS = from daily_checkins readiness score
- HOLD STABILITY = best avg/shot across series (approximation from ACC if Polar connected; else score-based)
- MENTAL SCORE = psychology composite score

Data Source: [performance_summary](#) table + [session_series](#) + [psychology_scores](#)

1.27 Full Session Report Screen

Screen Name: [SessionReportScreen](#)

Route: [/session/report](#)

Purpose: Detailed session performance comparison with athlete historical average.

Component	Type	Notes
Back arrow	Icon button	
"HamsaTech"	Supertitle	
"Shooting / Training"	Heading	
"Daily session performance and average comparison"	Subtitle	
HOLD STABILITY card	MetricCard	"6.5" + "● +0.3 vs avg"
CONSISTENCY card	MetricCard	"62%" + "● -9%"
MENTAL FOCUS card	MetricCard	"55" + "● steady focus"
LATEST SCORE card	MetricCard	"6.2" + "● +0.12"
"Today vs Athlete Average" card	Card	"Today: 6.2" vs "Period avg: 6.03" side by side

Component	Type	Notes
"Latest Session Score Pattern" card	Card	Series bubbles: "65" / "58" circles, "Session score, focus and comparison" caption
"Continue to Insights"	Primary button	→ InsightsScreen

1.28 Insights Screen

Screen Name: **InsightsScreen**

Route: **/session/insights**

Purpose: Psychology + physiology analysis after session completion.

Component	Type	Notes
Back arrow	Icon button	
"Today's Sessions"	AppBar title	
"Astra Performance"	Supertitle	Teal
"Insights"	Heading	
"Psychology, physiology and score-vs-HR trust"	Subtitle	
SOCIAL card	MetricCard	"60" + "● functional"
AROUSAL card	MetricCard	"50" + "● conditioning" (orange)
DECISION card	MetricCard	"90" + "● elite"
FOCUS card	MetricCard	"80" + "● elite"
"Physiology Explained" card	Card	HR badge + HRV badge + ACC badge with text explanations (Polar-connected only)
"Performance Recommendation" card	Card	Bold headline + body paragraph + bullet points
"Score vs HR Trust" chart	Card	Line chart: Score trust (teal) / Heart rate (red) / Threshold (orange dashed), legend below
"Save and Continue"	Primary button	→ Back to Home or next flow step

Polar-Not-Connected State ("Physiology Insights" card):

Shows: "Polar not connected. Connect Polar to unlock BPM, HRV, and physiology insights." + "Connect Polar Device" button

Psychology Dimension Labels (visible):

- Social, Arousal, Decision, Focus (from profile psychology_scores)

Data Source:

- psychology_scores (latest)
- hr_stream (session aggregated — if Polar connected)
- ai_insights (performance recommendation)
- performance_summary

API Call:

```
GET /rest/v1/ai_insights?athlete_id=eq.ATH001&session_id=eq.<uuid>&order=priority.desc&limit=3
GET /rest/v1/psychology_scores?athlete_id=eq.<uuid>&session_id=eq.<uuid>
```

1.29 Profile Screen

Screen Name: ProfileScreen

Route: /profile

Purpose: Athlete identity, performance stats, psychology scores, and goal management.

Component	Type	Notes
Back arrow	Icon button	
"Profile"	AppBar title	
"Astra Performance"	Supertitle	Teal
"Athlete Profile"	Heading	
"Identity, coach, psychology and score history"	Subtitle	
Avatar	CircleAvatar	"A" placeholder (no photo set)
Athlete name "Athlete"	Text	Bold
"Coach — Not linked"	Caption	
Streak counter	Badge	"🔥 1 streak" (with-Polar) / "🔥 0 streak" (without-Polar)
Rank badge	Badge	"Lvl 1 rank"
Attribute chips	Row	"Right dominant" / "Intermediate" / "Training"
BEST AVG card	MetricCard	"—" (30 days)
PERIOD AVG card	MetricCard	"—" (2 sessions)
BEST SERIES card	MetricCard	"—" (all time)

Component	Type	Notes
LAST SESSION card	MetricCard	"2" sessions total
"Quick Psychology Scores"	Section heading	
Score bars	List	Social 72 / Arousal 61 / Decision 80 / Focus 82 / Recovery 66 (colored progress bars)
"Goals" section	Section	
"Edit Goals →"	Text link	→ Goals editing screen (Needs backend confirmation)
Short-term (30 Days)	GoalRow	"Not set yet" + "0%" progress
Long-term (6 Months)	GoalRow	"Not set yet" + "0%" progress
Dream Goal	GoalRow	Star icon + "Not set yet"
"Sign Out"	Destructive button	Red, full width → logs out, clears tokens, → /welcome

Navigation (bottom nav):

Profile tab is active. Other tabs: Home / Sessions / Insight

Sign Out Behavior:

- 1. Clear [auth_token](#), [refresh_token](#) from SharedPreferences
- 2. Clear [onboarding_complete](#) (do NOT clear — user stays onboarded)
- 3. Navigate → [/welcome](#)
- 4. (Non-blocking) Supabase Auth signOut call

1.30 Sessions List Screen

Screen Name: [SessionsScreen](#)

Route: [/sessions](#)

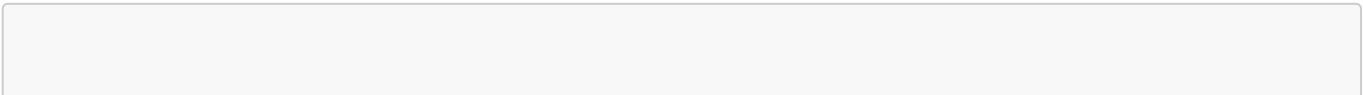
Purpose: Browse session history. (**Needs backend confirmation** — screenshots show post-session summary on this tab, not a list)

Inferred from bottom nav label "Sessions":

- May show session history list OR redirect to last session summary
- **Needs backend confirmation** for full list vs. latest-only behavior

2. Screen-by-Screen Flow Map

2.1 First-Time User Flow (With Polar)





2.2 First-Time User Flow (Without Polar)

```

... (same through PolarPermissionsScreen)
├── "I don't have a polar yet" → AthleteHomeScreen (Polar-not-
connected state)
└──

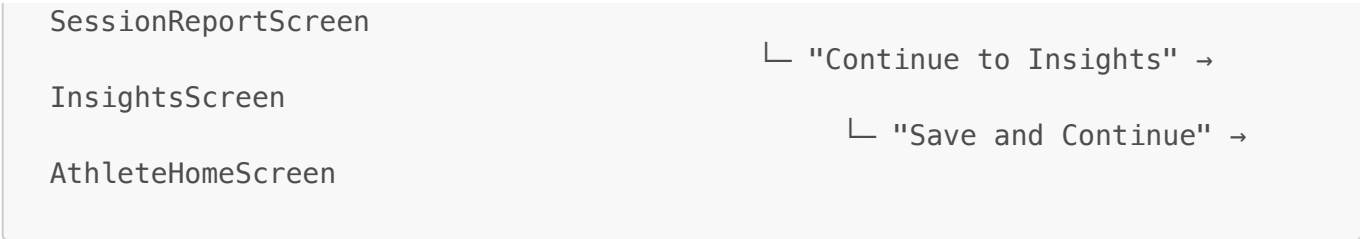
```

2.3 Session Flow (Full — With Polar)

```

AthleteHomeScreen
├── "Start Training Session" → DailyCheckinScreen
│   ├── "Save Check-in and Continue" → SessionSetupScreen
│   │   ├── "Begin Ritual" → PreSessionRitualScreen
│   │   │   ├── (ritual complete) → LiveTrainingScreen (Polar HR
streaming active)
│   │   │   └── "End Session" → ScoreEntryScreen
│   │   │       ├── (all series entered) → "Continue to Summary"
│   │   └── "Save & View Summary" →
│   └── "Save & View Full Report" →
└── ReflectionScreen
    └── SessionSummaryScreen
        └──

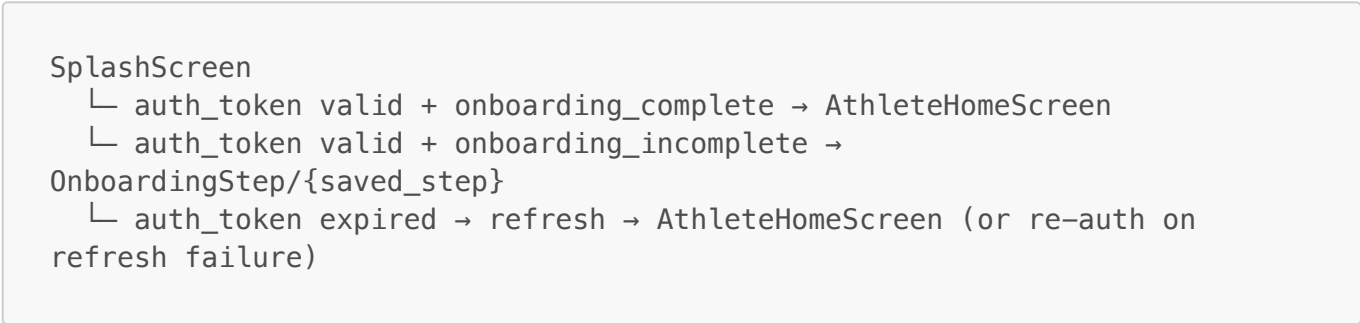
```



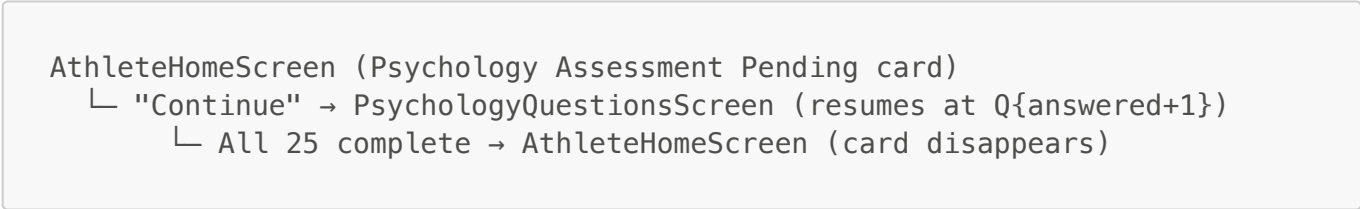
2.4 Session Flow (Without Polar)

- (Same as above, but:)
- LiveTrainingScreen: no HR display; may be skipped or shows timer only
 - InsightsScreen: Physiology card shows "Polar not connected" upsell

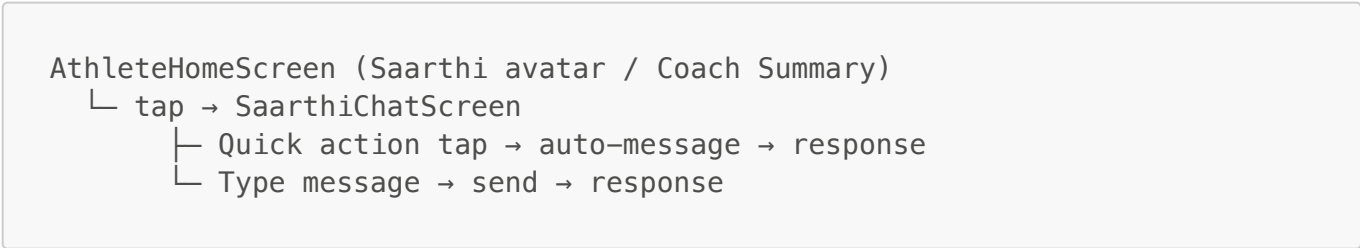
2.5 Returning User Flow



2.6 Psychology Assessment Flow (Post-Onboarding)



2.7 Saarthi Chat Flow



3. API Contract Requirements (Per Screen)

3.1 Auth Endpoints

Action	Method	Endpoint	Auth Required
Send OTP	POST	/auth/v1/otp	None (anon key)

Action	Method	Endpoint	Auth Required
Verify OTP	POST	/auth/v1/verify	None (anon key)
Refresh token	POST	/auth/v1/token?grant_type=refresh_token	refresh_token
Sign out	POST	/auth/v1/logout	access_token

Send OTP Request/Response:

```
// Request
{ "phone": "+919876543210" }

// Response 200
{ "message_id": "...", "phone": "+919876543210" }

// Error 429
{ "error": "rate_limit_exceeded", "message": "..." }
```

Verify OTP Request/Response:

```
// Request
{ "phone": "+919876543210", "token": "123456", "type": "sms" }

// Response 200
{
  "access_token": "<jwt>",
  "token_type": "bearer",
  "expires_in": 900,
  "refresh_token": "<opaque>",
  "user": { "id": "<uuid>", "phone": "+919876543210" }
}

// Error 400 (wrong OTP)
{ "error": "otp_expired", "message": "Token has expired or is invalid" }
```

Retry Logic:

- OTP send: no auto-retry (manual resend via "Didn't get a code?")
- Verify: 3 attempts before locking

Offline Behavior: OTP flows require network. Show "No internet connection" toast and disable submit button.

3.2 Onboarding Endpoints

Create Athlete Profile (Step 4 completion):


```
// POST /rest/v1/athletes
// Headers: Prefer: return=representation, Authorization: Bearer <jwt>
{
  "user_id": "<App_Users.id>",
  "athlete_id": "ATH{auto_generated}",
  "full_name": "Jane Doe",
  "gender": "female",
  "sport": "shooting",
  "discipline": "air_pistol"
}

// Response 201
[{ "id": "<uuid>", "athlete_id": "ATH001", ... }]
```

Create Athlete Details:

```
// POST /rest/v1/athlete_details
{
  "athlete_id": "<athletes.id>",
  "dominant_hand": null,
  "goals": ["Score 560 in 30 days", "Compete at nationals 590+"]
}
```

Failure States:

- 409 Conflict (duplicate athlete_id) → regenerate ID + retry
- 401 Unauthorized → re-authenticate
- 503 Network error → store locally, sync on next app open

Offline Behavior: Store all onboarding data in SharedPreferences. Sync to Supabase on next successful network call.

3.3 Session Lifecycle Endpoints

Create Session:

```
// POST /rest/v1/App_Sessions
// Headers: Prefer: return=representation
{
  "athlete_id": "ATH001",
  "session_type": "training",
  "range_type": "paper",
  "planned_shots": 60,
  "start_time": "2026-05-19T05:00:00Z",
  "status": "active"
}
```

```
// Response 201
[{ "id": "<uuid>", "status": "active", ... }]
```

Pause/Resume Session:

```
// PATCH /rest/v1/App_Sessions?id=eq.<session_id>
{ "status": "paused" } // or "active"
// Response 204
```

Complete Session:

```
// PATCH /rest/v1/App_Sessions?id=eq.<session_id>
{ "status": "completed", "end_time": "2026-05-19T07:00:00Z" }
// Response 204
```

Failure States:

- Session create fails → sessionId = null, continue locally, retry on next open
- Session complete fails → retry silently up to 3 times with exponential backoff (2s, 4s, 8s)

Offline Behavior: Session data (setup, scores) stored in SharedPreferences. Sync when network available.

3.4 Telemetry Endpoints

Bulk HR Insert:

```
// POST /rest/v1/hr_stream
// Headers: Prefer: return=minimal
[
  {
    "time": "2026-05-19T05:01:00Z",
    "session_id": "<uuid>",
    "athlete_id": "ATH001",
    "bpm": 72,
    "rr_interval": 833,
    "source": "polar_h10"
  }
]
// Response 201 (empty body due to return=minimal)
```

Flush Strategy:

- Every 10 seconds OR 500 rows, whichever first
- On Pause: flush immediately
- On End Session: flush all remaining

Failure Handling:

- Network error during flush → buffer retained, retry next flush interval
 - Max buffer size 2000 rows → oldest entries dropped with warning log
-

3.5 Performance Summary Endpoint

```
// POST /rest/v1/performance_summary
// Headers: Prefer: return=representation
{
  "athlete_id": "ATH001",
  "session_id": "<uuid>",
  "total_score": 123.0,
  "max_possible": 218.0,
  "avg_per_shot": 6.15,
  "score_variance": 0.89,
  "consistency_pct": 62.0,
  "performance_rating": 6,
  "focus_level": "moderate"
}
// Response 201
[{ "id": "<uuid>", ... }]
```

Calculation (done client-side before POST):

```
total_score = sum(series_totals)
max_possible = series_count * shots_per_series * 10.9
avg_per_shot = total_score / total_shots
performance_rating = clamp(round((total_score / max_possible) * 10), 1, 10)
focus_level = rating >= 7 ? "high" : rating >= 5 ? "moderate" : "low"
consistency_pct = (1 - (std_dev(per_shot_avgs) / mean(per_shot_avgs))) * 100
```

3.6 Check-in Endpoint

```
// POST /rest/v1/daily_checkins
// Headers: Prefer: return=representation
{
  "athlete_id": "<uuid>",
  "checkin_date": "2026-05-19",
  "mood": 8,
  "energy": 5,
  "sleep_quality": 7,
  "sleep_hours": 7.0,
  "stress_level": null,
}
```

```
"soreness": null,  
"notes": "Focused,Motivated"  
}
```

Idempotency: UPSERT on (`athlete_id`, `checkin_date`) unique constraint.

```
// PATCH /rest/v1/daily_checkins?athlete_id=eq.<uuid>&checkin_date=eq.  
<today>  
// if record already exists for today
```

3.7 Saarthi Edge Function

```
// POST /functions/v1/saarthi-chat  
// Headers: Authorization: Bearer <user_jwt>  
{  
  "athlete_id": "<uuid>",  
  "session_id": "<uuid>",  
  "message": "How do I manage pre-competition anxiety?",  
  "context_mode": "general"  
}  
  
// Response 200  
{  
  "reply": "Pre-competition anxiety is your body's preparation signal...",  
  "suggestions": [  
    "Try 4-7-8 breathing 10 minutes before",  
    "Review your pre-shot routine cues"  
  ],  
  "mood_flag": "high_anxiety"  
}  
  
// Response 500 (AI error)  
{  
  "error": "upstream_error",  
  "reply": "I'm having trouble connecting right now. Please try again."  
}
```

Context Mode Values: `"pre_session"` | `"post_session"` | `"general"`

Offline Behavior: Queue message locally. Show "Message will be sent when online." Retry when network restores.

3.8 Dashboard RPC

```
// POST /rest/v1/rpc/get_dashboard_data
// Headers: Authorization: Bearer <user_jwt>
{ "p_athlete_id": "ATH001" }

// Response 200
{
  "avg_score_7d": 513.4,
  "consistency_7d": 88.3,
  "sessions_this_week": 4,
  "readiness_score": 76,
  "trend": "improving"
}

// Note: RPC existence unconfirmed. Caller catches all errors silently.
```

4. Database Mapping (Per Screen)

Screen	Table READ	Table WRITE	Required Fields
Splash	—	—	—
Welcome	—	—	—
Sign Up	—	—	—
Phone Entry	—	—	—
OTP Verify	App_Users (created by Supabase Auth)	App_Users	auth_uid, phone
Onboarding 1–3	—	SharedPreferences only	—
Onboarding 4	—	athletes, athlete_details	full_name, athlete_id, discipline
Psychology Qs	psychology_questions	psychology_responses	question_id, response_value
Polar Permissions	—	—	—
Polar Scan	—	—	—
Polar Connected	—	polar_devices	athlete_id, device_id
Baseline Capture	—	SharedPreferences	baseline_hr
Baseline Result	athlete_details	athlete_details (PATCH)	resting_hr

Screen	Table READ	Table WRITE	Required Fields
Onboarding Complete	—	SharedPrefs	onboarding_complete = true
Athlete Home	daily_checkins, ai_insights, performance_summary	—	—
Saarthi Chat	athlete_chat (last 20)	athlete_chat	athlete_id, role, content
Daily Check-in	daily_checkins (today)	daily_checkins	athlete_id, checkin_date, mood, energy, sleep_quality
Session Setup	—	App_Sessions	athlete_id, session_type, start_time
Pre-Session Ritual	—	—	—
Live Training	—	hr_stream (bulk), App_Sessions (status)	time, session_id, bpm
Score Entry	—	performance_summary	session_id, total_score
Reflection	—	session_reflections, App_Sessions (complete)	session_id, emotion_rating
Session Summary	performance_summary, session_series	—	—
Session Report	performance_summary	—	—
Insights	ai_insights, psychology_scores, hr_stream	ai_insights (mark read)	—
Profile	athletes, athlete_details, psychology_scores	—	—

5. State Management Flow

5.1 BLoC Responsibilities

BLoC	Feature	States	Events
------	---------	--------	--------

BLoC	Feature	States	Events
AuthBloc	auth	AuthInitial, AuthLoading, Authenticated, Unauthenticated	CheckAuth, SendOtp, VerifyOtp, SignOut
OnboardingBloc	onboarding	StepState(1–4), OnboardingComplete	StepSubmitted, StepBack
SessionBloc	session	SessionInitial, SessionLoading, SessionStarted, SessionPaused, SessionCompleted	SessionStartRequested, SessionPauseRequested, SessionCompleteRequested
CheckinBloc	checkin	CheckinInitial, CheckinLoaded, CheckinSaved	LoadCheckin, SaveCheckin
ScoreEntryBloc	score_entry	ScoreEntryInitial, SeriesInProgress, AllSeriesEntered	SaveSeriesScore, UndoLastEntry
DashboardBloc	dashboard	DashboardLoading, DashboardLoaded, DashboardError	LoadDashboard
PolarBloc	polar	PolarDisconnected, PolarScanning, PolarConnected, PolarStreaming, PolarError	ScanDevices, ConnectDevice, StartStreaming, StopStreaming
SaarthiBloc	saarthi	SaarthiIdle, SaarthiTyping, SaarthiResponseReceived, SaarthiError	SendMessage, LoadHistory
ProfileBloc	profile	ProfileLoading, ProfileLoaded	LoadProfile
InsightsBloc	insights/dashboard	InsightsLoading, InsightsLoaded	LoadInsights, MarkInsightRead

5.2 SharedPreferences Key Registry

Key	Type	Written By	Read By
auth_token	String	OtpVerifyScreen	All API calls
refresh_token	String	OtpVerifyScreen	AuthBloc refresh
onboarding_step	int	Each onboarding screen	SplashScreen
onboarding_complete	bool	OnboardingCompleteScreen	SplashScreen

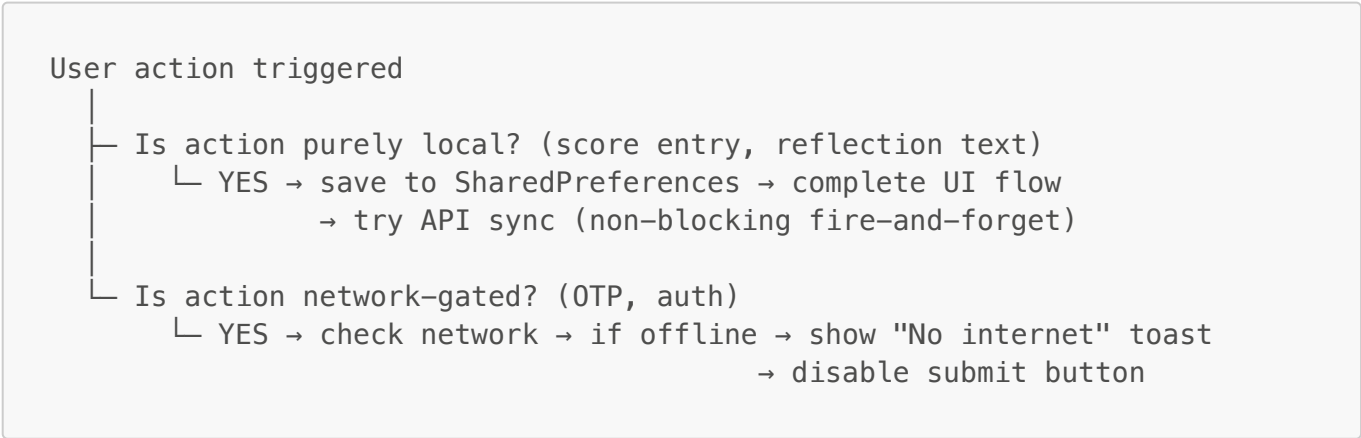
Key	Type	Written By	Read By
athlete_profile	JSON String	Onboarding Step 1–2	ProfileBloc
session_setup	JSON String	SessionSetupScreen	SessionBloc, ScoreEntryBloc
score_summary	JSON String	ScoreEntryBloc	SessionSummaryBloc
current_checkin	JSON String	CheckinBloc	DashboardBloc
baseline_scores	JSON String	Onboarding Step 3	ProfileBloc
baseline_hr	int	BaselineResultScreen	LiveTrainingScreen
polar_device_id	String	PolarConnectedScreen	PolarBloc
journal_entries	JSON String	ReflectionBloc	ProfileBloc

5.3 SessionMemory (In-Memory Only)

```
class SessionMemory {
  static String? sessionId;    // cleared on app restart
  static String? athleteId;   // set on login
}
```

Lifecycle: Set when session created in Supabase. Cleared on app cold-start. No persistence.

5.4 Offline-First Decision Tree



6. Polar BLE Flow

6.1 Full Pairing Sequence


```

Step 1: User taps "Continue & Allow" on PolarPermissionsScreen
└─ Request Android/iOS permissions:
    └─ Bluetooth Scan (Android 12+: BLUETOOTH_SCAN)
    └─ Bluetooth Connect (BLUETOOTH_CONNECT)
    └─ Location (for BLE scan on older Android)
    └─ Push Notifications
    └─ Microphone (optional)

Step 2: On permission grant → PolarScanScreen
└─ PolarBleApi.searchForDevice()
└─ List discovered devices (filter: model == "H10")
└─ User taps "Connect"

Step 3: PolarBleApi.connectToDevice(deviceId)
└─ Success → PolarConnectedScreen
└─ Failure → error card on PolarScanScreen

Step 4: PolarConnectedScreen
└─ Register device in polar_devices table (non-blocking)
└─ Save deviceId to SharedPreferences
└─ "Done Let's go" → BaselinePromptScreen

Step 5: Baseline Capture
└─ PolarBleApi.startHrStreaming(deviceId)
└─ Accumulate 60 samples
└─ Calculate mean BPM
└─ Stop HR stream
└─ Save result locally + PATCH athlete_details.resting_hr

Step 6: Session streaming (LiveTrainingScreen)
└─ PolarBleApi.startHrStreaming(deviceId) // 1 Hz
└─ Optional: PolarBleApi.startEcgStreaming(deviceId, settings) // 130
Hz
└─ Optional: PolarBleApi.startAccStreaming(deviceId, settings) // 200
Hz
└─ TelemetryBuffer.collect(sample)
└─ Every 10s: TelemetryBuffer.flush() → POST /rest/v1/hr_stream

```

6.2 Polar Sensor Settings (Dart)

```

// ECG
final ecgSettings = PolarSensorSetting({
  PolarSensorSetting.settingSampleRate: 130,
  PolarSensorSetting.settingResolution: 14,
});

// ACC
final accSettings = PolarSensorSetting({
  PolarSensorSetting.settingSampleRate: 200,
  PolarSensorSetting.settingResolution: 16,
});

```

```
PolarSensorSetting.settingRange: 8, // ±8G
});
```

6.3 Disconnect and Reconnection

On PolarBleApi deviceDisconnected event:

1. Log disconnect timestamp
2. Mark LiveTrainingScreen as "reconnecting..."
3. Wait 2 seconds (debounce)
4. Attempt reconnect: PolarBleApi.connectToDevice(savedDeviceId)
5. Retry schedule: 2s → 4s → 8s → 16s → 30s (max 5 attempts)
6. On success: resume streams, continue buffer
7. On permanent failure:
 - Stop streaming
 - Retain buffer in memory
 - Show "Polar disconnected" badge on LiveTrainingScreen
 - Allow session to continue without HR
 - Flush buffer when reconnected or on session end

6.4 Buffer Management

```
class TelemetryBuffer {
  static const _hrFlushInterval = Duration(seconds: 10);
  static const _ecgFlushInterval = Duration(seconds: 5);
  static const _maxBatchSize = 500; // rows per POST
  static const _maxBufferSize = 2000; // drop oldest beyond this

  List<Map<String, dynamic>> _hrBuffer = [];
  List<Map<String, dynamic>> _ecgBuffer = [];
  List<Map<String, dynamic>> _accBuffer = [];

  // Flush triggers: timer, maxBatchSize, session end
}
```

6.5 Live Training HR Display Logic

```
currentBpm = latest HR sample
baselineHr = SharedPreferences.getInt('baseline_hr')
delta = currentBpm - baselineHr
deltaLabel = delta > 0 ? "+$delta" : "$delta"
deltaColor = delta > 10 ? Colors.red : delta > 5 ? Colors.orange :
Colors.green

Display: "${currentBpm} bpm · Baseline ${baselineHr} ↑ ${deltaLabel}"
```

7. Saarthi AI Flow

7.1 Interaction Lifecycle

1. User opens SaarthiChatScreen

└ Load last 20 messages from athlete_chat (Supabase or local cache)

└ Display in chat bubble format

2. User types message or taps Quick Action

└ Append user message to chat (local, immediate)

└ Show typing indicator

3. POST /functions/v1/saarthi-chat

└ Edge Function builds context:

– Rolling 20 chat turns (from Redis or Supabase)

– Athlete snapshot (profile, discipline, experience)

– Today's check-in (mood, energy, sleep)

– Last session summary (score, consistency)

– Last 3 unread ai_insights

– Active session context (if session_id provided)

└ Calls Claude claude-sonnet-4-6 API

max_tokens: 1024, temperature: 0.7

└ Parses response

└ Inserts 2 rows into athlete_chat (user + assistant)

└ Updates Redis cache saarthi:context:{athlete_id}

└ Checks for mood_flag → queues insight generation if needed

4. Response received by Flutter

└ Append assistant reply to chat

└ Hide typing indicator

└ Show suggestions chips if present

5. Mood flag handling:

└ "high_anxiety" → flag for coach notification (Needs backend confirmation)

└ "low_motivation" → queue motivational insight generation

7.2 Context Mode Mapping

Trigger	context_mode	Additional context injected
Pre-session (from ritual screen)	pre_session	session setup params
Post-session (from insights)	post_session	session summary, scores
Home screen / any other	general	daily check-in

7.3 Insight Generation Pipeline

```

Trigger: session completed (ReflectionScreen saved)
→ Non-blocking Edge Function call: generate-insights

Edge Function steps:
1. Gather last 30 days of data:
    SELECT * FROM App_Sessions WHERE athlete_id = $1 ORDER BY created_at
DESC LIMIT 30
    SELECT * FROM performance_summary WHERE athlete_id = $1 ORDER BY
created_at DESC LIMIT 30
    SELECT * FROM daily_checkins WHERE athlete_id = $1 ORDER BY
checkin_date DESC LIMIT 30
    SELECT avg(bpm) FROM hr_stream WHERE athlete_id = $1 AND time > NOW()
- INTERVAL '30 days'

2. Build Claude prompt:
    "Analyze this athlete's last 30 days and identify 3 actionable
insights.
    Output JSON: [{type, title, body, priority}]"

3. Parse JSON response → INSERT INTO ai_insights (3 rows)
4. Mark old is_read=false insights as superseded (optional)
5. Return 201 to caller

Nightly job (pg_cron, Phase 2):
- Same pipeline for all athletes who had sessions in last 7 days
- Runs at 2 AM IST

```

7.4 Redis Cache Structure

```

saarthi:context:{athlete_id}
TTL: 24h
Value: {
  "chat_history": [...last 20 turns...],
  "athlete": { name, discipline, experience_years },
  "last_checkin": { mood, energy, sleep_hours },
  "last_session": { avg_per_shot, consistency_pct, total_score }
}

```

8. Coach / Admin Flow

From visible screenshots: No dedicated coach portal screens are visible in either PDF.

Inferred from Profile Screen:

- "Coach — Not linked" indicates a coach-linking feature exists
- Coaches table is defined in the schema
- `coach_feedback` table exists

Needs backend confirmation:

- Coach web portal (separate from mobile app?)
- Coach linking mechanism (invite code? QR code? search?)
- Coach dashboard (view multiple athletes)
- Coach feedback submission endpoint

From architecture.md RLS policies:

```
CREATE POLICY coach_read_sessions ON "App_Sessions"
FOR SELECT
USING (
  athlete_id IN (
    SELECT a.athlete_id FROM athletes a
    JOIN athlete_details ad ON ad.athlete_id = a.id
    WHERE ad.coach_id = (
      SELECT id FROM coaches WHERE user_id = auth.uid()
    )
  )
);
```

This confirms coach access is scoped but the UI is not yet visible.

9. Offline Behavior & Sync Strategy

9.1 Offline-Safe Operations (Local First)

Operation	Local Storage	Sync Trigger
Onboarding data entry	SharedPreferences	On each step completion
Daily check-in	SharedPreferences	Immediate non-blocking POST
Session setup	SharedPreferences	Session start
Score entry (all series)	SharedPreferences	After all series saved
Reflection	SharedPreferences	On save tap
Psychology responses	SharedPreferences	Per answer

9.2 Network-Required Operations

Operation	Reason
OTP send/verify	Auth is server-side
Saarthi chat	LLM call — no local fallback
AI insights fetch	Requires server generation

9.3 Sync Queue (Future Enhancement)

For Phase 2: implement a local **SyncQueue** that:

- 1. Enqueues failed API calls with payload
- 2. Retries on **Connectivity** restored event
- 3. Deduplicates by idempotency key

10. Error States & Edge Cases

10.1 Error State Catalogue

Screen	Error Condition	UI Treatment
OTP Verify	Wrong code	Inline error text below input
OTP Verify	Code expired	Toast + "Resend" link
Polar Scan	No devices found	Warning card (yellow border)
Polar Scan	BLE disabled	System prompt to enable BT
Baseline Capture	Polar disconnects	Error overlay + retry button
Live Training	Polar disconnects mid-session	"Polar disconnected" badge, session continues
Live Training	Network lost	Telemetry buffered locally
Score Entry	Score > max_per_series	Validation message "Score cannot exceed {max}"
API generic	401 Unauthorized	Auto-refresh token → retry once → re-login
API generic	503 / timeout	Silent for non-blocking calls; toast for blocking calls
Saarthi	Edge Function error	"I'm having trouble. Please try again." in chat

10.2 Empty States

Screen	Empty Condition	UI Treatment
Athlete Home	No check-in today	Readiness/Recovery/Stress show "—" or 0
Athlete Home	No sessions	"Start your first session" prompt
Profile	No sessions logged	"—" in BEST AVG, PERIOD AVG, BEST SERIES
Profile	No coach linked	"Coach — Not linked" (link action Needs backend confirmation)

Screen	Empty Condition	UI Treatment
Profile	Goals not set	"Not set yet" + "0%" progress bars
Insights	No AI insights	"Complete more sessions to unlock insights" (Needs backend confirmation)
Session History	No sessions	Empty list state (Needs backend confirmation)
Saarthi	No chat history	Quick actions grid (current implementation)

10.3 Uncertain / Needs Confirmation Items

#	Item	Where
1	Full discipline dropdown options	Onboarding Step 2
2	Carousel slides 2 and 3 content	WelcomeScreen
3	"Log In Here" flow — separate from Sign Up or same?	WelcomeScreen
4	"What if my number changes?" action	PhoneEntryScreen
5	Psychology question bank (all 25 questions, dimensions)	PsychologyQuestionsScreen
6	Pre-Session Ritual screen content/duration	PreSessionRitualScreen
7	Live Training without Polar — HR card shown or hidden?	LiveTrainingScreen
8	Electronic range score entry flow	ScoreEntryScreen
9	Sessions tab — list view vs. latest summary	SessionsScreen
10	Notification bell action	AthleteHomeScreen
11	Athlete ID auto-generation strategy (ATH001, ATH002...)	athletes table
12	Coach linking UX (invite code / search)	ProfileScreen
13	Push notification setup (APNs/FCM integration)	Phase 2
14	Stress/Soreness capture (present in schema, not visible in checkin screenshots)	DailyCheckinScreen
15	Score vs HR Trust chart — real-time or post-session computed?	InsightsScreen
16	<code>get_dashboard_data</code> RPC — does it exist?	AthleteHomeScreen
17	Dream Goal field — where is it set?	ProfileScreen
18	Short-term/Long-term goal progress calculation	ProfileScreen
19	Streak calculation logic (1-day streaks seen)	ProfileScreen
20	"Save & View Full Report" — does it generate a PDF?	SessionSummaryScreen

Appendix A — Screen Route Registry

Route	Screen	Auth Required
/	SplashScreen	No
/welcome	WelcomeScreen	No
/signup	SignUpScreen	No
/phone-entry	PhoneEntryScreen	No
/otp-verify	OtpVerificationScreen	No
/onboarding/step/1	OnboardingStep1Screen	Yes
/onboarding/step/2	OnboardingStep2Screen	Yes
/onboarding/step/3	OnboardingStep3Screen	Yes
/onboarding/step/4	OnboardingStep4Screen	Yes
/onboarding/complete	OnboardingCompleteScreen	Yes
/psychology/questions	PsychologyQuestionsScreen	Yes
/polar/permissions	PolarPermissionsScreen	Yes
/polar/scan	PolarScanScreen	Yes
/polar/connected	PolarConnectedScreen	Yes
/polar/baseline/prompt	BaselinePromptScreen	Yes
/polar/baseline/capturing	BaselineCapturingScreen	Yes
/polar/baseline/result	BaselineResultScreen	Yes
/home	AthleteHomeScreen	Yes
/saarathi	SaarathiChatScreen	Yes
/checkin	DailyCheckinScreen	Yes
/session/setup	SessionSetupScreen	Yes
/session/ritual	PreSessionRitualScreen	Yes
/session/live	LiveTrainingScreen	Yes
/session/score	ScoreEntryScreen	Yes
/session/reflection	ReflectionScreen	Yes
/session/summary	SessionSummaryScreen	Yes
/session/report	SessionReportScreen	Yes
/session/insights	InsightsScreen	Yes
/sessions	SessionsScreen	Yes

Route	Screen	Auth Required
/profile	ProfileScreen	Yes

Appendix B — Psychology Dimensions Observed

From Profile screen Quick Psychology Scores and Insights screen:

Dimension	Profile Bar Color	Insights Card
Social	Green	SOCIAL
Arousal	Orange	AROUSAL
Decision	Teal/dark	DECISION
Focus	Blue/purple	FOCUS
Recovery	Red	RECOVERY

Note: Onboarding psychology questions reference "focus, confidence, anxiety, motivation, resilience" per architecture.md, but the displayed dimensions in Profile/Insights use a different set (Social, Arousal, Decision, Focus, Recovery). **Needs backend confirmation** — alignment between question dimensions and display dimensions.

Appendix C — Metric Labels Mapping

Display Label	Source	Computation
READINESS	daily_checkins	Weighted: sleep × 0.30 + energy × 0.25 + mood × 0.20 + (10-stress) × 0.15 + (10-soreness) × 0.10, × 10
RECOVERY	daily_checkins	(sleep_hours/8)×40 + ((10-soreness)/10)×30 + ((10-stress)/10)×30
STRESS	daily_checkins	stress_level directly (Needs backend confirmation — not in checkin UI)
MENTAL STATE	psychology_scores	Derived label from composite_score
PERFORMANCE	performance_summary	performance_rating (1–10)
HOLD STABILITY	acc_stream or performance	Best avg/shot across series
MENTAL SCORE	psychology_scores	composite_score
CONSISTENCY	performance_summary	consistency_pct
SOCIAL	psychology_scores	social dimension score
AROUSAL	psychology_scores	arousal dimension score

Display Label	Source	Computation
DECISION	psychology_scores	decision dimension score
FOCUS	psychology_scores	focus_score
RECOVERY (psych)	psychology_scores	resilience/recovery dimension score

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